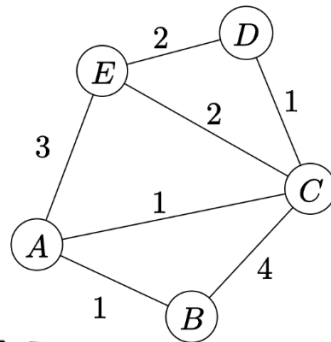


cut(V) is set of all edges
 $\rightarrow$ 1 end pt. in V
 $\rightarrow$ other in complement of V

Common Data for Questions (1) and (2)
 Consider the following weighted graph:



sum of wts of the crossing edges

$$A-B \rightarrow 1$$

$$A-C \rightarrow 1$$

$$A-E \rightarrow 3$$

$$B-C \rightarrow 4$$

$$C-D \rightarrow 1$$

$$C-E \rightarrow 2$$

$$D-E \rightarrow 2$$

$$V \sim \{A, B, C, D, E\}$$

1) Among the cuts given below, which of them has the smallest cumulative weight?

1 point

- ☐ {A, B, C} $\Rightarrow 3 + 1 + 2 \Rightarrow 6$
- ☐ {C, D, E} $\Rightarrow 1 + 4 + 3 \Rightarrow 8$
- ☐ {A, C, E} $\Rightarrow 1 + 4 + 1 + 2 \Rightarrow 8$
- ☐ {A, E} $\Rightarrow 1 + 1 + 2 + 2 \Rightarrow 6$
- ☒ {A} $\Rightarrow 1 + 1 + 3 \Rightarrow 5$

Yes, the answer is correct.

2) Among the following select all cuts that have a cumulative weight of 8.

1 point

- ☐ {A, B, C}
- ☒ {C, D, E}
- ☒ {A, C, E}
- ☐ {A, E}
- ☐ {A}

Yes, the answer is correct.

3) Which of the following are true for the Laplacian matrix of a weighted undirected graph, $G = (V, E, W)$, with positive edge weights? Note that w_{ij} is the entry in the i^{th} row, j^{th} column of W . Also note that $w_{ij} = 0$ if $(i, j) \notin E$. 2 points

- ☒ L is positive semi-definite. $\rightarrow$ all eigen values of $L \geq 0$
- ☐ L is positive definite.
- ☒ The smallest eigenvalue of L is zero.
- ☒ $x^T L x = \sum_{(i,j) \in E} w_{ij} (x_i - x_j)^2$, where $x = (x_1, \dots, x_n)$.
- ☐ L is invertible and its pseudo-inverse is just the inverse.
- ☒ If $v_i = v_j$ for all i, j with $1 \leq i, j \leq n$, then $v = (v_1, \dots, v_n)$ is an eigenvector of L .

Yes, the answer is correct.

$$G = (V, E, W)$$

$$L = D - W$$

$$D_{ij} = \sum_j w_{ij}$$

$L = D - W$
 for every node,
 $d = 2w_1 + w_2$
 $= \frac{2e+1}{e^2}$
 all diag. entries of L is d



$$w_{ij} = e^{-\frac{\|x_i - x_j\|^2}{2}}$$

$$\|x_i - x_j\|^2 = 2$$

$$w_{ij} = e^{-1} = \frac{1}{e} \quad (\text{dist} = \sqrt{2})$$

$$w_2 = \frac{1}{e^2} \quad (\text{dist} = 2)$$

$$L = D - W$$

Common Data for Questions (4) to (8)

Consider the following dataset in $\mathbb{R}^2$:

$$x_1 = (0, 1)$$

$$x_2 = (1, 0)$$

$$x_3 = (0, -1)$$

$$x_4 = (-1, 0)$$

Instantiate a weighted graph on this dataset using the weighting function given in the lectures with $\sigma = 1$. Compute the Laplacian, L , of this graph. L_{ij} is the element in the i^{th} row and j^{th} column, with $1 \leq i, j \leq 4$.

4) What is L_{22} ?

☐ $\frac{1}{e}$

☒ $\frac{2e+1}{e^2}$

☐ $\frac{1}{e^2}$

☐ $-\frac{1}{e}$

☐ $-\frac{1}{e^2}$

$$L_{ii} = d_i = \sum_{j \neq i} w_{ij}$$

$$L_{22} = 2 \cdot \frac{1}{e} + \frac{1}{e^2} = \frac{2e+1}{e^2}$$

1 point

Yes, the answer is correct.

5) What is L_{13} ?

☐ $\frac{1}{e}$

☐ $\frac{2e+1}{e^2}$

☐ $\frac{1}{e^2}$

☐ $-\frac{1}{e}$

☒ $-\frac{1}{e^2}$

off-diagonal $L_{ij} = -w_{ij} \quad (i \neq j)$

$x_1 = (0, 1)$
 $x_3 = (0, -1)$

$$w_{13} = \frac{1}{e^2}$$

$$L_{13} = -\frac{1}{e^2}$$

$$d = \frac{2e+1}{e^2} \quad w_1 = \frac{1}{e}$$

$$w_2 = -\frac{1}{e^2}$$

1 point

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$-\frac{1}{e^2}$$

6) What is L_{41} ?

☐ $\frac{1}{e}$

☐ $\frac{2e+1}{e^2}$

☐ $\frac{1}{e^2}$

☒ $-\frac{1}{e}$

$$L_{ij} = -w_{ij} \quad (i \neq j)$$

$x_4 = (-1, 0)$
 $x_1 = (0, 1)$

$$w_{41} = \frac{1}{e}$$

$$L_{41} = -\frac{1}{e}$$

$$L = \begin{bmatrix} d & -w_1 & -w_2 & -w_1 \\ -w_1 & d & -w_1 & -w_2 \\ -w_2 & -w_1 & d & -w_1 \\ -w_1 & -w_2 & -w_1 & d \end{bmatrix}$$

1 point

7) Find the smallest non-zero eigenvalue of L and enter the closest integer as your answer. You can use code for this.

1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 1

1 point

8) Retain the first two eigenvectors, v_1, v_2 , corresponding to the two smallest non-zero eigenvalues, λ_1, λ_2 , with $\lambda_1 \leq \lambda_2$. Use the rows as embeddings after normalization. Which of the following is true regarding the position of points in the plane, ignoring the indices of the points? You can use code for this.

☒ The data-points and the embeddings coincide.

☐ The embeddings and data-points do not coincide and occupy distinct locations.

☐ Cannot comment on this with the given information.

Yes, the answer is correct.

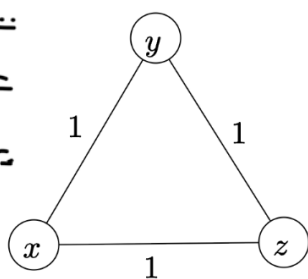
$$V = \begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

V & original they both have the same circle shape.

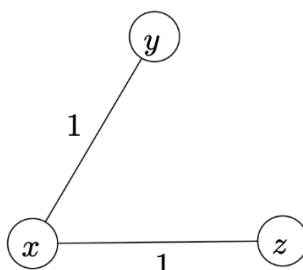
Common Data for Questions (9) to (15)

Consider the weighted graph G on the left and its sparsification H on the right:

$$\begin{aligned} \deg(x) &= 2 \\ \deg(y) &= 2 \\ \deg(z) &= 2 \end{aligned}$$



$$G = (V, E)$$



$$H = (V, F)$$

$$\begin{aligned} L &= D - W \\ D_{ii} &= \sum_j w_{ij} \text{ (degree of } i) \\ L_{ij} &= -w_{ij} \text{ (} i \neq j) \\ L_{ii} &= D_{ii} \end{aligned}$$

9) Compute L_G , the Laplacian of G . The first three rows(columns) correspond to x, y, z , in that order.

1 point

☐ $\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

☒ $\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

$$L_G = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

10) Compute L_H , the Laplacian of H . The first three rows(columns) correspond to x, y, z , in that order.

1 point

☐ $\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

☒ $\begin{bmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$$\begin{aligned} \deg(x) &= 2 \\ \deg(y) &= 1 \\ \deg(z) &= 1 \end{aligned}$$

$$L_H = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

11) Compute the quadratic form $\theta^T L_G \theta$, where $\theta = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

1 point

☒ $(x-y)^2 + (y-z)^2 + (z-x)^2$

☐ $x^2 + y^2 + z^2 + xy + yz + zx$

☐ $(x-y)^2 + (y-z)^2$

$$\theta^T L_G \theta = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{1 \times 3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}_{3 \times 3} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1}$$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 2x - y - z & -x + 2y - z & -x - y + 2z \end{bmatrix}_{1 \times 3}$$

$$\theta^T L_G \theta = \sum_{(i,j)} w_{ij} (\theta_i - \theta_j)^2$$

$$G \Rightarrow xy, yz, zx \text{ with } w = 1$$

$$(x-y)^2 + (y-z)^2 + (z-x)^2$$

$$H = xy, xz$$

$$(x-y)^2 + (x-z)^2$$

12) Compute the quadratic form $\theta^T L_H \theta$, where $\theta = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

1 point

- ☐ $(x-y)^2 + (y-z)^2 + (z-x)^2$
- ☐ $x^2 + y^2 + z^2 + xy + yz + zx$
- ☐ $(x-y)^2 + (y-z)^2$
- ☒ $(x-y)^2 + (z-x)^2$
- ☐ $x^2 + y^2 + z^2 - xy - yz - zx$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$(x-y)^2 + (z-x)^2$

official

13) Under what conditions is H an ϵ -spectral approximation of G ?

1 point

- ☐ $(1-\epsilon)\theta^T L_H \theta \leq \theta^T L_G \theta \leq (1+\epsilon)\theta^T L_H \theta$, for all $\theta \in \mathbb{R}^3$
- ☒ $(1-\epsilon)\theta^T L_G \theta \leq \theta^T L_H \theta \leq (1+\epsilon)\theta^T L_G \theta$, for all $\theta \in \mathbb{R}^3$
- ☐ $(1-\epsilon)\theta^T L_G \theta \leq \theta^T L_H \theta \leq (1+\epsilon)\theta^T L_H \theta$, for all $\theta \in \mathbb{R}^3$
- ☐ $(1-\epsilon)\theta^T L_G \theta \leq \theta^T L_G \theta \leq (1+\epsilon)\theta^T L_H \theta$, for all $\theta \in \mathbb{R}^3$

$$(1-\epsilon) \theta^T L_G \theta \leq \theta^T L_H \theta \leq (1+\epsilon) \theta^T L_G \theta$$

Yes, the answer is correct.

14) In this and the next question, we will arrive at a range for ϵ for which H is an ϵ -spectral approximation of G , if such an approximation exists. First we focus on the RHS of the inequality obtained in the previous question.

The right-hand side of the inequality is satisfied for what range of ϵ ? Choose the best possible (widest) range.

- ☐ $0 < \epsilon < \frac{2}{3}$
- ☐ $\frac{2}{3} < \epsilon < 1$
- ☒ $0 < \epsilon < 1$
- ☐ $0 < \epsilon < \frac{1}{3}$
- ☐ There is no such ϵ .

$$\theta^T L_H \theta \leq (1+\epsilon) \theta^T L_G \theta$$

$$(x-y)^2 + (z-x)^2 \leq (1+\epsilon) [(x-y)^2 + (y-z)^2 + (z-x)^2]$$

$$(y-z)^2 \geq 0$$

$$1 \leq 1+\epsilon \Rightarrow 0 < \epsilon < 1$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$0 < \epsilon < 1$

15) Now repeat the same procedure for the LHS of the inequality. This should allow you to answer the final question. For what range of ϵ is H an ϵ -spectral approximation of G ?

4 points

- ☐ $0 < \epsilon < \frac{2}{3}$
- ☒ $\frac{2}{3} < \epsilon < 1$
- ☐ $0 < \epsilon < 1$
- ☐ $0 < \epsilon < \frac{1}{3}$
- ☐ There is no such ϵ .

$$(1-\epsilon) \theta^T L_G \theta \leq \theta^T L_H \theta$$

$$\frac{(x-y)^2 + (z-x)^2}{(x-y)^2 + (y-z)^2 + (z-x)^2} \geq 1-\epsilon$$

solve this inequality

$$\frac{2}{3} < \epsilon < 1$$